MILITARY SPECIFICATION

DC TO DC INVERTER, ELECTRONIC TIMER MODULE

This specification is approved for use by the U.S Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packing of one type of Inverter, DC to DC, Electronic Timer Module for the XM-122 firing device receiver.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-105	-	Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-202	-	Test Methods for Electrical and Electronic Equipment
MIL-STD-750	-	Test Methods for Semiconductor Devices
MIL-STD-883	-	Test Methods and Procedures for Microelectronics
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment

DRAWINGS (See 6.5)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9312632 - Inverter, DC to DC, Electronic Timer Module

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assembly shall comply with all requirements specified in drawing 9312632 and associated drawings and with all requirements specified in applicable specifications and standards.

Microcircuit devices shall, as a minimum, pass the Class B screening and lot quality conformance requirements of Methods 5004 and 5005 of MIL-STD-883.

Semiconductor devices shall, as a minimum, be Jan TX devices or equivalent. When Jan TX devices are not available, the devices shall be screened and burned in to the degree comparable to Jan TX types as defined in MIL-STD-750.

3.3 Vibration. The module shall not show evidence of damage or failure when tested in accordance with Method 201 of MIL-STD-202.

3.4 Humidity. The module shall not show evidence of damage when tested as specified in Method 103, of MIL-STD-202, Condition B.

3.5 Temperature. The module shall not show evidence of damage when tested per MIL-STD-202, Method 107, Condition A.

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3.6 Electrical performance characteristics

3.6.1 +3.25 volt turn-on time. The voltage at pin E3, with pin E3 loaded with an eighteen thousand two hundred plus or minus ten percent (18,200 ± 10%) ohm resistor in parallel with one microfarad plus or minus ten percent (1.0 ± 10%) microfarad capacitor, shall rise to plus three and twenty-five hundredths volts direct current plus or minus twenty-five hundredths volts direct current (+3.25 VDC ± .25 VDC) within ten (10) seconds after application of the input voltage to pin E2 when tested as specified in 4.5.1.1.

3.6.2 +55 volt turn-on time. The voltage at pin E4 shall be less than plus five tenths volts direct current (+.5 VDC) a minimum of four (4) minutes after the application of the input voltage to pin E2 and shall rise to plus fifty-five volts direct current plus or minus five volts direct current (+55 VDC ± 5 VDC) at some time between four (4) and eight (8) minutes after the application of the input voltage to pin E2. The above requirements shall be met when the module is tested as specified in 4.5.1.2.

3.6.3 +55 volt maximum. The voltage at pin E4 shall not exceed plus sixty volts (+60 volts) at any time between four (4) and eight (8) minutes after the application of the input voltage to pin E2 when tested as specified in 4.5.1.3.

3.6.4 +55 volt output characteristic. The output on pin E4 shall be plus fifty-five volts direct current plus or minus five volts direct current (+55 VDC ± 5 VDC) when the input voltage to pin E2 is varied between plus one and twenty hundredths volt direct current and plus one and fifty hundredths volt direct current (+1.20 VDC and 1.50 VDC) when tested as specified in 4.5.1.4.

3.6.5 End of life/+3.25 volts. The plus fifty-five volt (+55 V) source shall discharge to less than three volts direct current (+3 VDC) and remain discharged whenever the voltage source to pin E2 is decreased below the "end of life" voltage. The "end of life" voltage shall be less than plus one and twenty hundredths volt direct current (+1.20 VDC). For voltages at pin E2 greater than "end of life" voltage, the voltage output at pin E3 shall remain regulated as per the requirements of 3.6.11 (below). The above requirements must be met when tested as specified in 4.5.1.5.

3.6.6 End of life/+55 volts. The plus fifty-five volt (+55 V) source shall discharge to less than three volts direct current (+3.0 VDC) and remain discharged whenever the voltage source to pin E2 is decreased below the "end of life" voltage. For voltages at pin E2 greater than the "end of life" voltage, the voltage output at pin E4 shall be plus fifty-five plus or minus five volts direct current (+55 ± 5 VDC). The above requirements must be met when tested as specified in 4.5.1.6.

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3.6.7 Reset characteristic. The output voltage at pin E4 shall be internally discharged to less than plus three volts direct current (+3.0 VDC) within five (5) seconds whenever the voltage source at pin E2 is removed five tenths (.5) second or longer. The module shall meet the requirements of 3.6.1 and 3.6.2 upon the reapplication of the voltage source to pin E2. The above requirements must be met when tested as specified in 4.5.1.7.

3.6.8 +55 volt capacitance characteristic. The capacitance between pins E4 and E1 shall be two hundred and thirty microfarads plus or minus ten percent (230 ufd ± 10%) at room temperature and two hundred and thirty microfarads plus twenty percent minus ten percent (230 ufd +20%, -10%) at temperature extremes, when tested as specified in 4.5.1.8.

3.6.9 +55 volt load characteristic. The module shall be capable of operation after pin E4 is discharged into a load of five plus or minus ten percent (5 ± 10%) ohms. These requirements must be met when tested as specified in 4.5.1.9.

3.6.10 Charging characteristics of the +55 volt source. The maximum charging current to the internal plus fifty five volt (+55V) energy source at pin E4 shall be two and zero tenths milliamperes (2.0 ma). The maximum voltage charging rate of the energy source shall not exceed twenty volts per second (20 V/s). The above requirements shall be met when tested as specified in 4.5.1.10.

3.6.11 +3.25 volt regulation. The voltage output at pin E3 shall be within the range of plus three and zero hundredths volts direct current and plus three and fifty hundredths volts direct current (+3.00 VDC and +3.50 VDC) when the input voltage on pin E2 is varied between plus one and twenty hundredths volts direct current and plus one and fifty hundredths volts direct current (+1.20 VDC and +1.50 VDC) under all load conditions and during charging of the 55 volt internal source when tested as specified in 4.5.1.11.

3.6.12 +3.25 volt load transients/regulation. The voltage output at pin E3 shall be within the range of plus three and zero hundredths volts direct current and plus three and fifty hundredths volts direct current (+3.00 VDC and +3.50 VDC) when the input voltage on pin E2 is varied between plus one and twenty hundredths volts direct current and plus one and fifty hundredths volts direct current (+1.20 VDC and 1.50 VDC) and when the load on pin E3 is switched from the eighteen thousand two hundred plus or minus two percent (18,200 ± 2%) ohm load to the five hundred plus or minus two percent (500 ±2%) ohm load and vice versa when tested as specified in 4.5.1.12.

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3.6.13 +3.25 volt output ripple. The voltage output at pin E3 shall not have more than one hundred fifty millivolts (150 mv) peak to peak ripple under all load conditions and during charging of the internal 55 volt source. The above requirements are to be met when tested as specified in 4.5.1.13.

3.6.14 +3.25 volt load transient/output ripple. The voltage output at pin E3 shall not have more than one hundred fifty millivolts (150 mv) peak to peak ripple when the load on pin E3 is switched from the eighteen thousand, two hundred plus or minus two percent (18,200 ± 2%) ohm load to the five hundred plus or minus two percent (500 ± 2%) ohm load and vice versa when tested as specified in 4.5.1.14.

3.6.15 +3.25 volt output loading characteristics. The output at pin E3 shall be capable of supplying continuous current into a five hundred plus or minus two percent (500 ± 2%) resistive load and meet the requirements of 3.6.11 when tested as specified in 4.5.1.15.

3.6.16 Battery current drain. The following requirements shall be met when tested as specified in 4.5.1.16.

3.6.16.1 Battery current while charging. The current supplied to pin E2 during the charging of the plus fifty five volt (+55 V) energy source shall not exceed eighty milliamperes (80 ma) when the load on pin E3 is an eighteen thousand two hundred plus or minus two percent (18,200 ± 2%) ohm resistive load.

3.6.16.2 Battery current while charging/heavy load. The current supplied to pin E2 during the charging mode of the plus fifty five volt (+55 V) energy source shall not exceed eighty milliamperes (80 ma) when the load on pin E3 is five hundred plus or minus two percent (500 ± 2%) ohms resistive load.

3.6.16.3 Battery current during light load. The current supplied to pin E2 after the time-out and the energy source charging cycles are complete shall not exceed three and five tenths milliamperes (3.5 ma) when pin E3 is loaded with an eighteen thousand two hundred plus or minus two percent (18,200 ± 2%) resistive load.

3.6.16.4 Battery current during heavy load. The current supplied to pin E2 after the time-out and energy source charging cycles are complete shall not exceed eighty and zero tenths milliamperes (80.0 ma) when pin E3 is loaded with a five hundred plus or minus two percent (500 ± 2%) ohm resistive load.

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3.6.17 Voltage variations. The module shall meet the requirements of 3.6.4, 3.6.11 and 3.6.16 when the voltage source applied to pin E2 is varied to the specified limits when tested as specified in 4.5.1.17.

3.6.18 Temperature variations. The module shall meet the requirements of 3.6.1 through 3.6.14 over the temperature environment range of minus thirty-two degrees centigrade (-32°C) to plus fifty-two degrees centigrade (+52°C) when tested as specified in 4.5.1.18.

3.6.19 Low Battery Voltage Characteristics. The plus fifty-five volt (+55V) source shall remain in a discharged condition whenever the input voltage applied at pin #2 is less than "END OF LIFE". This requirement shall be tested as specified in 4.5.1.19.

3.7 Rework or repair. Any module failing to comply with all of the test requirements and undergoes rework or repair shall be tested as specified in 4.5.2.

3.8 Test temperature. Unless specified otherwise, all testing shall be performed at a temperature of twenty-five degrees centigrade plus or minus five degrees centigrade (25°C ± 5°C).

3.9 Potting compound

3.9.1 Process. The process for storage, mixing and dispensing shall be performed in a manner that will not cause degradation of the compound. The process will be tested as specified in 4.5.3.1.

3.9.2 Cure. The potted module shall assure proper curing when tested as specified in 4.5.3.2.

3.10 Data requirements. Data shall be recorded and retained for all tests required by this specification and as specified in 4.5.4.

3.11 Workmanship. The module shall be free from dirt, chips, grease, flux, rust or other foreign matter that may impair proper functioning. The cleaning methods used shall not be injurious to any part or assembly nor shall the parts or assembly be contaminated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection.

b. Quality conformance inspections.

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of twenty-five (25) modules.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table 1 specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I: FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63144A(AR)

PARAGRAPH	TITLE Inverter, DC to DC, Electronic Timer Module		SHEET 1 OF 3		DRAWING NUMBER 9312632
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inverter, DC to DC, Electronic Timer Module (before potting) (dwg 9312632)				
	Examination for defects	25		3.2	4.4.2.1
	Tests				
	+3.25 Volt Turn-on Time	25(a)		3.6.1	4.5.1.1
	+55 Volt Turn-on Time	25(a)		3.6.2	4.5.1.2
	+55 Volt Maximum	25(a)		3.6.3	4.5.1.3
	+55 Volt Output Characteristic	25(a)		3.6.4	4.5.1.4
	End of Life/+3.25 Volts	25(a)		3.6.5	4.5.1.5
	End of Life/+55 Volts	25(a)		3.6.6	4.5.1.6
	Reset Characteristic	25(a)		3.6.7	4.5.1.7
	+55 Volt Output Capacitance Characteristic	25(a)		3.6.8	4.5.1.8
	+55 Volt Load Characteristic	25(a)		3.6.9	4.5.1.9
	Charging Characterisitics of the +55 Volt Source	25(a)		3.6.10	4.5.1.10
	+3.25 Volt Regulation	25(a)		3.6.11	4.5.1.11
	+3.25 Volt Load Transient/Regulation	25(a)		3.6.12	4.5.1.12
	+3.25 Volt Output Ripple	25(a)		3.6.13	4.5.1.13
	+3.25 Volt Load Transient/Output Ripple	25(a)		3.6.14	4.5.1.14
	+3.25 Output Loading Characteristics	25(a)		3.6.15	4.5.1.15
	Battery Current While Charging	25(a)		3.6.16.1	4.5.1.16
	Battery Current While Charging/Heavy Load	25(a)		3.6.16.2	4.5.1.16
	Battery Current During Light Load	25(a)		3.6.16.3	4.5.1.16
	Battery Current During Heavy Load	25(a)		3.6.16.4	4.5.1.16

NOTES:

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TABLE I: FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63144A(AR)

PARAGRAPH	TITLE Inverter, DC to DC, Electronic Timer Module		SHEET 2 OF 3		DRAWING NUMBER 9312632
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inverter, DC to DC, Electronic Timer Module (after potting) (dwg 9312632)				
	Examination for defects	25		3.2	4.4.2.2
	Mechanical Test				
	Vibration	25 (b)		3.3	
	Temperature cycling	25 (b)		3.5	
	Humidity	25 (b)		3.4	
	Processes				
	Potting Compound Process	25		3.9.1	4.5.3.1
	Potting Compound Cured	25		3.9.2	4.5.3.2
	Electrical Test				
	+3.25 Volt Turn-on Time	25 (c)		3.6.1	4.5.1.1
	+55 Volt Turn-on Time	25 (c)		3.6.2	4.5.1.2
	+55 Volt Maximum	25 (c)		3.6.3	4.5.1.3
	+55 Volt Output Characteristic	25 (c)		3.6.4	4.5.1.4
	End of Life/+3.25 Volts	25 (c)		3.6.5	4.5.1.5
	End of Life/+55 Volts	25 (c)		3.6.6	4.5.1.6
	Reset Characteristic	25 (c)		3.6.7	4.5.1.7
	+55 Volt Output Capacitance Characteristic	25 (c)		3.6.8	4.5.1.8
	+55 Volt Load Characteristic	25 (c)		3.6.9	4.5.1.9
	Charging Characteristics of the +55 Volt Source	25 (c)		3.6.10	4.5.1.10
	+3.25 Volt Regulation	25 (c)		3.6.11	4.5.1.11
	+3.25 Volt Load Transient/Regulation	25 (c)		3.6.12	4.5.1.12

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TABLE I: FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63144A(AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
	Inverter, DC to DC, Electronic Timer Module				9312632
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	+3.25 Volt Output Ripple	25(c)		3.6.13	4.5.1.13
	+3.25 Volt Load Transient/Output Ripple	25(c)		3.6.14	4.5.1.14
	+3.25 Output Loading Characteristics	25(c)		3.6.15	4.5.1.15
	Battery Current While Charging	25(c)		3.6.16.1	4.5.1.16
	Battery Current While Charging/Heavy Load	25(c)		3.6.16.2	4.5.1.16
	Battery Current During Light Load	25(c)		3.6.16.3	4.5.1.16
	Battery Current During Heavy Load	25(c)		3.6.16.4	4.5.1.16
	Voltage Variations	25(d)		3.6.17	4.5.1.17
	Temperature Variations	25(d)		3.6.18	4.5.1.18
	Low Battery Voltage Characteristics	25(c)		3.6.19	4.5.1.19
	NOTES: (a) Tests shall be performed at +25°C + 5°C. (b) Mechanical tests shall be performed prior to conducting post pot electrical test and in the sequence shown. (c) Tests shall be performed at +25°C + 5°C. Readings shall be within 10% of pre-pot readings from (a) above (d) Voltage and temperature variations shall be performed after the tests required by note (c).				

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the classification of defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63144A(AR)

PARAGRAPH	TITLE				DRAWING NUMBER
4..4.2.1	DC to DC Inverter/Electronic Timer Module (Before Potting)		1 SHEET 1 OF		9312632
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Components incorrectly installed, broken or missing		0.40%	3.2	Visual
102	Printed circuit board damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.11	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63144A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.2	Inverter, DC to DC, Electronic Timer Module				9312632
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of connector		0.40%	3.2	Gage
102	Maximum height		0.40%	3.2	Gage
103	Maximum length		0.40%	3.2	Gage
104	Maximum width		0.40%	3.2	Gage
105	Location of guidehole		0.40%	3.2	Gage
106	Location of mounting bracket		0.40%	3.2	Gage
107	Potting compound process correct		0.40%	3.9.1	4.5.3.1
108	Potting compound cured		0.40%	3.9.2	4.5.3.2
109	Potting not covering circuit board		100%	3.2	Visual
110	+55 Volt turn-on time		100%	3.6.2	4.5.1.2
111	+55 Volt output characteristics		100%	3.6.4	4.5.1.4
112	End of life +3.25 volts		100%	3.6.5	4.5.1.5
113	Reset characteristic		100%	3.6.7	4.5.1.7
114	+55 volt output capacity characteristic		100%	3.6.8	4.5.1.8
115	+55 volt load characteristic		100%	3.6.9	4.5.1.9
116	+3.25 volt regulation		100%	3.6.11	4.5.1.11
117	+3.25 volt output ripple		100%	3.6.13	4.5.1.13
118	+3.25 output loading characteristics		100%	3.6.15	4.5.1.15
119	Battery current while charging		100%	3.6.16.1	4.5.1.16
120	Battery current while charging/heavy load		100%	3.6.16.2	4.5.1.16
121	Battery current during light load		100%	3.6.16.3	4.5.1.16
122	Battery current during heavy load		100%	3.6.16.4	4.5.1.16
123	Voltage variations		100%	3.6.17	4.5.1.17
124	Temperature variations		100%	3.6.18	4.5.1.18
125	Low battery voltage characteristic		100%	3.6.19	4.5.1.19

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63144A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Inverter, DC to DC, Electronic Timer Module				9312632
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
126	Charging characteristics of the +55 volt source		100%	3.6.10	4.5.1.10
Minor					
201	Terminals missing or damaged		0.65%	3.2	Visual
202	Evidence of poor workmanship		100%	3.11	Visual

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference Inspection Method column in the tables of Paragraphs 4.4.2.1, 4.4.2.2 and 4.4.2.3. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures

4.5.1 Electrical test procedures. The test equipment shall be accurate to within two percent (2%) of the nominal value of the parameter being measured unless otherwise specified. The nominal input voltage applied to E2 shall be plus one and fifty hundredths plus or minus ten hundredths volts direct current (+1.50 ± .10VDC). All voltage and current measurements are referenced to pin E1 (ground potential). Unless otherwise specified, a load of eighteen thousand, two hundred plus or minus two percent (18,200 ± 2%) ohms in parallel with a one microfarad plus or minus ten percent (1.0 uf ± 10%) capacitor shall be connected to pin E3. All test fixtures, equipment, and procedures shall be approved by the procuring activity.

4.5.1.1 +3.25 volt turn-on time. The module shall meet the turn-on time requirements for the output voltage at pin E3. The rise time at pin E3 shall be measured to an accuracy of one (1) second. Any module failing to comply with the specified requirements shall be classed defective (non-destructive test).

4.5.1.2 +55 volt turn-on time. The module shall meet the turn-on time requirements for the output voltage at pin E4 from plus two volts direct current (+2.0 VDC) to greater than plus fifty volts direct current (+50.0 VDC). Any module failing to comply with the specified requirements shall be classed as defective (non-destructive test).

4.5.1.3 +55 volt maximum. The module shall meet the maximum voltage requirement for the output voltage at pin E4. Any module failing to comply with the specified requirements shall be classed as defective (non-destructive test).

4.5.1.4 +55 volt output characteristic. The module shall meet the plus fifty five volt (+55V) output characteristic requirement for the output voltage at pin E4. Any module failing to comply with the specified requirements shall be classed as defective (non-destructive test).

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4.5.1.5 End of life/+3.25 volts. The module shall meet the "end of life" requirement for pin E3 loaded with a five hundred ohm plus or minus two percent (500 ohm ± 2%) resistor applied after the voltage on pin E4 has reached plus fifty-five volts plus or minus five volts direct current (+55 ± 5 VDC). The nominal input voltage shall be applied to pin E2. After the voltage on pin E4 has risen to plus fifty-five volts plus or minus five volts direct current (+55 VDC ± 5VDC), the voltage on pin E2 shall be slowly decreased until the "end of life" condition has occurred. The "end of life" voltage at pin E2 shall be recorded. The voltage at pin E3 shall remain plus three and twenty five hundredths plus or minus twenty five hundredths volts direct current (+3.25 ± .25 VDC) for voltages at pin E2 that are greater than "end of life". Any module failing to meet the specified requirements shall be classed as defective (non-destructive test).

4.5.1.6 End of life/+55 volts. The module shall meet the "end of life" requirement with pin E3 loaded with a five hundred ohm plus or minus two percent (500 ± 2%) resistor applied after the voltage on pin E4 has reached plus fifty-five volts plus or minus five volts direct current (+55 ± 5 VDC). The nominal input voltage shall be applied to pin E2. After the voltage on pin E4 has risen to plus fifty-five plus or minus five volts direct current (+55 ± 5 VDC), the voltage on pin E2 shall be slowly decreased until the "end of life" condition has occurred. The voltage at pin E4 shall remain plus fifty-five plus or minus 5 volts direct current (+55 ± 5VDC) for voltages at pin E2 that are greater than "end of life". Any module failing to meet the specified requirements shall be classed as defective (non-destructive test).

4.5.1.7 Reset characteristic. Apply the nominal input voltage to pin E2. After the turn-on requirements of the voltages at pin E3 and E4 have been met, the input voltage to E2 shall be interrupted for a time greater than five tenths second (.5 sec). Measure the voltage and the voltage discharge time at pin E4. Any module failing to meet the specified requirement shall be classed as defective (non-destructive test).

4.5.1.8 +55 volt output capacitance characteristic. The capacitance between pins E4 and E1 of the module shall be measured with a capacitance meter. No input voltage shall be applied to pin E2. Any module failing to meet all of the required requirements shall be classed as defective (non-destructive test).

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4.5.1.9 +55 volt load characteristic. Apply the nominal input voltage to pin E2. After the turn-on requirement of the voltage at pin E4 has been met, momentarily connect five ohms plus or minus ten percent (5 ohms ± 10%) between pins E4 and E1. Allow the voltage on pin E4 to be recharged to fifty-five volts direct current plus or munis five volts (+55 ± 5 VDC). Again momentarily connect the resistor. Any module failing to meet the required specification shall be classed as defective (non-destructive test).

4.5.1.10 Charging characteristic of the +55 volt source. Connect a current meter between pins E4 and E1 with no input voltage applied. Apply the nominal input voltage to pin E2. Measure the current when charging occurs. Remove the current meter and allow capacitor to charge. After the charging cycle of the module is complete, discharge the capacitive energy source at pin E4 by momentarily connecting a five plus or minus ten percent (5 ± 10%) ohm resistor between pins E1 and E4, and allow the capacitor to recharge. Determine maximum charging rate characteristic of the source with an oscilloscope or timing device. Any module failing to meet all of the specified requirements shall be classed as defective (non-destructive test).

4.5.1.11 +3.25 voltage regulation. Connect a five hundred ohm plus or minus two percent (500 ohm ± 2%) resistor between pins E3 and E1. Apply the input voltage to pins E2 and vary it over the required range of plus one and twenty hundredths volts direct current and plus one and fifty hundredths volts direct current (+1.20 VDC and +1.50 VDC). Measure the direct current (DC) voltage on pin E3. Repeat the above test with an eighteen thousand two hundred ohm plus or minus two percent (18,200 ohm ± 2%) resistor. Any module failing to meet the specified requirement shall be classed as defective (non-destructive test).

4.5.1.12 +3.25 volt load transient/regulation. Observe on an oscilloscope the voltage on pin E3 when the load on pin E3 is being switched from the heavy load condition to the light load condition and vice versa. Any module failing to meet the specified requirement shall be classed as defective (non-destructive test).

4.5.1.13 +3.25 volt output ripple. Connect a five hundred ohm plus or minus two percent (500 ohm ± 2%) resistor between pins E3 and E1. Apply the input voltage to pin E2 and vary it over the required voltage range of plus one and twenty hundredths volts direct current and plus one and fifty hundredths volts direct current (+1.20 VDC and +1.50 VDC). Measure the ripple on the pin E3. Repeat the above test with an eighteen thousand two hundred ohm plus or minus two percent (18,200 ohm ± 2%) resistor.

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4.5.1.14 +3.25 volt load transient/ripple. Observe on a oscilloscope the voltage on pin E3 when the load on pin E3 is being switched from the heavy load condition to the light load condition and vice versa. Any module failing to meet the specified requirement shall be classed as defective (non-destructive test).

4.5.1.15 +3.25 output loading characteristic. +3.25 volt output current with heavy load is met by performance of the test specified in 4.5.1.11. Any module failing to meet all of the specified requirements shall be classed as defective (non-destructive test).

4.5.1.16 Battery current drain. Connect a current meter between the input voltage source and pin E2. Measure the currents under the following module conditions:

a. The +55 volt source is being charged and a load resistor of eighteen thousand two hundred ohms plus or minus two percent (18,200 ohms ± 2%) is connected between pins E3 and E1.

b. The module is in the charging mode and a load resistor of five hundred plus or minus two percent (500 ± 2%) ohms is connected between pins E3 and E1.

c. After the charging of the +55 volt source is complete with a load of eighteen thousand two hundred ohms plus or minus two percent (18,200 ± 2%).

d. The +55 volt is charged and a load resistor of five hundred ohms plus or minus two percent (500 ohms ± 2%) is connected between pins E3 and E1.

Any module failing to meet the specified requirements shall be classed as defective (non-destructive test).

4.5.1.17 Voltage variations. The module shall be tested for specified requirements with the input voltage at pin E2 varied to the specified limits. The tests to be conducted are specified in Table II.

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TABLE II

TEST VOLTAGE	FIRST ARTICLE INSPECTION TESTS	QUALITY CONFORMANCE INSPECTION TESTS
1.50 VDC	4.5.1.1 4.5.1.2 4.5.1.3 4.5.1.4 4.5.1.8 4.5.1.9 4.5.1.10 4.5.1.11 4.5.1.12 4.5.1.13 4.5.1.14	4.5.1.4 4.5.1.11 4.5.1.16
1.20 VDC	4.5.1.1 4.5.1.2 4.5.1.3 4.5.1.4 4.5.1.8 4.5.1.9 4.5.1.10 4.5.1.11 4.5.1.12 4.5.1.13 4.5.1.14	4.5.1.4 4.5.1.11 4.5.1.16
Less than End of Life	4.5.1.5 4.5.1.6 4.5.1.7	

4.5.1.18 Temperature variations. The module shall be tested for specified requirements over the temperature range. Readings shall be taken at temperatures of minus thirty-two degrees centigrade (-32°C) maximum and plus fifty-two degrees centigrade (+52°C) minimum. The voltage variations and the tests to be conducted are specified in Table II. (Units shall be soaked a minimum of one hour at temperature extremes.)

4.5.1.19 Low Battery Voltage Characteristics. Apply an input voltage of plus one volt direct current plus or minus one tenth volt direct current (+1 VDC ± .1 VDC) to pin E2 with a load on pin E3 of eighteen thousand two hundred plus or minus two percent (18,200 ± 2%) ohms. The input current into pin E2 shall exceed thirty milliamperes (30 ma) direct current (an input current greater than 30 ma indicates the discharge condition). Any module failing to comply with the specified requirement shall be classified as defective (non-destructive test).

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4.5.2 Rework or retest. Any Inverter, DC to DC, Electronic Timer failing to meet all of the requirements and is reworked or repaired shall be subject to all of the test performances of 4.5.1.1 through 4.5.1.14 before being classed acceptable. Any module failing to comply with all of the requirements after rework shall be classed defective. All rework or repair shall be documented as to cause of rejection and the work performed. The rejected data shall accompany the data for acceptance of the module (non-destructive test).

4.5.3 Potting compound

4.5.3.1 Potting compound process. The potting compound shall be stored, mixed and dispensed in a manner approved by the Government QA activity specified in 6.4 before use by the contractor. Failure to comply with these requirements shall classify the material used by the contractor as non-conforming and it shall not be acceptable for use as a deliverable item.

4.5.3.2 Potting compound cured. The potted module shall be tested for potting compound curing using Government approved test equipment and a cure schedule. The cure schedule shall be developed by the contractor and submitted to the QA activity specified in 6.4 before use by the contractor. Failure to comply with all of the specified requirements shall classify the sample as non-conforming and it shall not be acceptable for use as a deliverable item.

4.5.4 Data requirements. The contractor shall submit a quality inspection test, demonstration and evaluation report in accordance with Data Item Description DI-R-1724 (tailored) giving all detailed inspection and test results.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Each inverter/electronic timer module shall be preserved and packaged in accordance with MIL-STD-1188A.

5.2 Packing. The inverter, preserved and packaged in 5.1, shall be packed for shipment in accordance with MIL-STD-1188A.

5.3 Marking. Marking of the carton shall comply with MIL-STD-1188A.

5.4 Shipping. When Inverter, DC to DC, Electronic Timer of more than one lot are shipped together, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

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6. NOTES

6.1 Intended use. The assemblies covered by this specification are intended for use as an Inverter, DC to DC, Electronic Timer for the XM-122 firing device receiver.

6.2 Ordering data. See Section 6.1 and 6.2 of MIL-A-48078.

6.3 Submission of inspection equipment designs for approval (See MIL-A-48078). Submit equipment designs as required to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

6.4 Distribution of ammunition data cards, inspection results and test data. Distribution of ammunition data cards, inspection C and test results shall be submitted to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAA-M, Dover, New Jersey 07801.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-63144A DC TO DC INVERTER, ELECTRONIC TIMER MODULE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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